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A
Project Report
On

ZEN-PARK: An AI-Enabled Smart Parking Management System

submitted as partial fulfillment for the award of

BACHELOR OF TECHNOLOGY DEGREE

SESSION 2024-25

in

COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

By

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Under the supervision of

Dr. Priya Singh

KIET Group of Institutions, Ghaziabad

Affiliated to
Dr. A.P.J. Abdul Kalam Technical University, Lucknow
(Formerly UPTU)

February 2026

DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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Date: 20 February 2026

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CERTIFICATE

3

This is to certify that Project Report entitled "ZEN-PARK: An AI-Enabled Smart Parking Management System" which is submitted by Prabhav Tiwari in partial fulfillment of the requirement for the award of degree B. Tech. in Department of CSE (AIML) of Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

66

Dr. Priya Singh

(Asst. Prof)

Dr. Rekha Kashyap

Dean – CSE(AI/AI&ML)

Date: 20 February 2026

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Last but not the least, we acknowledge our friends for their contribution to the completion of the project.

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ABSTRACT

Parking bottlenecks have become a significant problem due to rapid urbanization and rising vehicle density. Traditional parking management systems, including manual ticketing and RFID-based access control, suffer from inefficiencies, security vulnerabilities, and limited scalability. To address these challenges, this project presents ZenPark, an Artificial Intelligence-enabled smart parking management system integrating real time vehicle detection, automatic license plate recognition, dynamic slot allocation, and multi-layered security mechanisms.

The proposed system utilizes YOLOv11 for high-accuracy vehicle detection and EasyOCR for license plate identification. A hybrid edge cloud architecture is implemented to optimize latency and scalability. The system incorporates infrared-based spoof detection and encrypted communication protocols to enhance security and protect against unauthorized access or adversarial manipulation.

The system was deployed in the institutional parking environment and evaluated using metrics such as detection accuracy, OCR accuracy, end-to-end latency, and scalability. Experimental results demonstrate that ZenPark achieves decent detection precision, reliable recognition accuracy, and sub 10 second average response time under optimum operating conditions.

The presented solution offers a scalable, cost-effective, and secure approach for smart parking management in institutional, commercial, and smart city environments. Future enhancements may include blockchain-based logging, federated learning integration, and smart city infrastructure connectivity.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
ANPR	Automatic Number Plate Recognition
OCR	Optical Character Recognition
IoT	Internet of Things
CNN	Convolutional Neural Network
CRNN	Convolutional Recurrent Neural Network
mAP	Mean Average Precision
FPS	Frames Per Second
TLS	Transport Layer Security
API	Application Programming Interface
JWT	JSON Web Token
FGSM	Fast Gradient Sign Method
PGD	Projected Gradient Descent
RTO	Regional Transport Office
ITS	Intelligent Transportation System
V2I	Vehicle-to-Infrastructure

CHAPTER 1
INTRODUCTION

1.1 INTRODUCTION

Citification has led to a rapid rise in the number of vehicles across cities worldwide. With growing population density and expanding commercial infrastructure, the want for efficient parking systems has become more crucial than ever. In metropolitan and semi-urban areas, a large portion of traffic bottlenecks are caused by vehicles searching for available parking spaces. Studies indicate that nearly 30% of urban traffic consists of drivers circling around in search of parking, leading to more fuel consumption, higher carbon emissions, and unnecessary delays.

In India, the problem is even more pronounced due to unplanned urban expansion, limited infrastructure, and inefficient parking management practices. Conventional parking systems commonly rely on manual ticketing, security guards, or basic RFID-based entry systems. While these methods provide partial or some automation, they do suffer from several limitations such as human error, fraud, ticket loss, lack of real time monitoring, and inability to dynamically allocate parking slots based on demand.

Mechanical parking structures and sensor-based smart parking systems have been introduced to address these issues. However, mechanical systems are expensive to construct and maintain, and IoT sensor-based systems require one sensor per parking slot, resulting in high installation and maintenance costs. Additionally, many existing smart parking implementations do not adequately address security concerns such as spoofing attacks, unauthorized access, or data breaches.

With advancements in Artificial Intelligence and Computer Vision, camera-based parking management systems have emerged as a promising alternative. Modern deep learning models are capable of detecting vehicles in real time and recognizing license plates with high

accuracy. These technologies reduce the need for physical sensors and enable centralized monitoring of multiple parking slots using a limited number of cameras.

63 Object detection models belonging to the YOLO (You Only Look Once) family have significantly improved and increased real time detection capabilities. The latest generation models provide enhanced accuracy, reduced latency, and improved robustness against environmental variations. Similarly, Optical Character Recognition (OCR) technologies allow automatic extraction of vehicle registration numbers, eliminating manual data entry and enabling seamless automation of parking operations.

4 Despite these advancements, most existing solutions focus on either vehicle detection or slot monitoring, but rarely integrate detection, license plate recognition, security mechanisms, and cloud-based management into unified architecture. Furthermore, security remains an overlooked aspect in many academic prototypes, making them vulnerable to spoofing attacks and adversarial manipulation.

To address these limitations, the present project introduces ZenPark, an AI-enabled smart 4 parking management system that integrates advanced vehicle detection, license plate recognition, dynamic slot allocation, and multi-layered security mechanisms within a cloud–59 edge hybrid architecture. The system intends to provide an efficient, scalable, and secure solution suitable for institutional campuses, commercial complexes, and smart city deployments.

This chapter introduces us of the parking problem, discusses the limitations of conventional systems, and outlines the motivation behind the development of our project ZenPark.

1.2 PROJECT DESCRIPTION

ZenPark is an Artificial Intelligence driven parking management system designed to automate vehicle entry, slot allocation, monitoring, and security validation. The system leverages deep learning based object detection, Optical Character Recognition (OCR), and cloud edge computing to provide a comprehensive parking management solution.

The primary objective of ZenPark is to reduce congestion caused by inefficient parking processes and to enhance operational transparency and security. The system replaces manual record-keeping with automated vehicle detection and digital logging, ensuring faster processing and reduced human dependency.

The overall workflow of ZenPark is as follows:

- Cameras installed at the parking entrance capture real-time video streams.
- The captured frames are processed using a deep learning-based vehicle detection model (YOLOv11) to identify vehicles.
- Once a vehicle is detected, the license plate characters are extracted.
- The extracted plate image is processed using EasyOCR to recognize the vehicle registration number.
- The recognized number is verified against the system database.
- If the vehicle is authorized, a parking slot is dynamically allocated.
- Entry logs are stored securely in the cloud database.
- If unauthorized access or suspicious behavior is detected, alerts are triggered.

The system adopts hybrid edge cloud architecture. Edge devices perform preliminary processing to reduce bandwidth consumption, while computationally intensive tasks such as large-scale inference and database management are handled in the cloud. This architecture ensures both low latency and high scalability.

One of the distinguishing features of ZenPark is its focus on security. The system integrates infrared based spoof detection to differentiate between genuine number plates and printed or replayed forgeries. Encrypted API communication, authentication mechanisms, and adversarial attack resistance are incorporated to ensure system reliability and data protection.

This project was implemented and evaluated in a controlled institutional parking environment. Performance metrics such as detection accuracy, OCR accuracy, latency, scalability, and spoof detection effectiveness were analyzed to validate system efficiency.

The key objectives of ZenPark are:

- To design a real time AI based vehicle detection system.
- To implement automatic license plate recognition for entry logging.
- To develop a dynamic slot allocation mechanism.
- To ensure secure communication and spoof resistance.
- To create a scalable and cost effective parking system.

It is important to clarify that ZenPark is designed as a custom enterprise level smart parking solution. The project does not propose a novel detection algorithm or new deep learning architecture. Instead, it focuses on the integration, optimization, and secure deployment of existing state-of-the-art models into a practical and scalable parking management system suitable for real world environments.

Scope of the Project:

The system is primarily designed for medium-scale parking environments such as educational institutions, office complexes, and gated communities. However, the architecture is modular and scalable, allowing extension to larger smart city infrastructures with minimal modifications.

Organization of the Report:

The remaining chapters of this report are organized as follows:

- Chapter 2 presents a detailed literature review of present parking systems and related technologies.
- Chapter 3 describes the presented methodology, system architecture, algorithms, and implementation details.
- Chapter 4 discusses experimental results, performance evaluation, and analysis.
- Chapter 5 concludes the project and outlines future enhancements.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview of Parking Management Systems

Parking management has evolved largely over the past few decades or so. With rapid citification and rising vehicle ownership, researchers and industries have continuously attempted to improve the efficacy of parking systems. Old parking systems were entirely manual, relying on attendants to record vehicle details and manage slot allocation. Over time, semi-automated and fully automated solutions emerged, incorporating RFID, IoT sensors, and computer vision technologies.

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The primary goal of modern parking research is to reduce congestion, lower the search time, enhance security, and improve overall user convenience. However, each technological generation has introduced both advantages and new limitations.

2.2 Manual and Ticket-Based Parking Systems

Manual parking systems are the most traditional form of parking management. In such systems, human attendants are responsible for issuing entry tickets, recording vehicle details, and collecting fees at exit points.

Advantages:

- Low initial installation cost
- Minimal technical infrastructure required
- Easy implementation

Limitations:

- Human error in record keeping
- Fraud and ticket duplication
- Long waiting time during peak hours

- No real-time slot visibility
- No automated tracking of vehicle entry and exit

Manual systems are still widely used in small-scale parking environments. However, they are inefficient for large institutions and commercial complexes where high vehicle inflow demands faster processing.

2.3 RFID-Based Parking Systems

Radio Frequency Identification (RFID) systems introduced automation into parking management. Vehicles are equipped with RFID tags, and scanners detect authorized entries automatically.

Advantages:

- Reduced manual intervention
- Faster gate operation
- Moderate scalability

Limitations:

- RFID tags can be cloned or stolen
- No visual verification of vehicles
- Additional cost per vehicle
- Vulnerable to spoofing and unauthorized duplication

While RFID systems improve operational efficiency compared to manual methods, they do not provide visual confirmation of vehicle identity and are susceptible to security breaches.

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2.4 IoT Sensor Based Smart Parking Systems

The next advancement involved Internet of Things (IoT) based systems. These systems deploy ultrasonic, magnetic, or infrared sensors in each parking slot to detect vehicle occupancy. Sensor data is transmitted to a central server, which updates slot availability in real time.

Working Principle:

- Each parking slot has a dedicated sensor.
- The sensor detects vehicle presence.
- Data is transmitted using wireless communication protocols such as MQTT or Wi-Fi.
- Slot availability is displayed via dashboards or mobile applications.

Advantages:

- Real-time slot-level monitoring
- Integration with mobile apps
- Improved parking space utilization
- Limitations:
- High installation cost (one sensor per slot)
- Maintenance complexity
- Sensor malfunction due to weather conditions
- Single-point failure risks

Although IoT-based systems provide accurate slot monitoring, large-scale deployment becomes expensive and maintenance-intensive.

2.5 Vision-Based Parking Systems

With rapid advancement of computer vision and deep learning, camera based parking systems gained popularity. Instead of installing sensors per slot, strategically placed cameras monitor multiple slots simultaneously.

Benefits of Vision-Based Systems:

- Reduced hardware dependency
- Centralized monitoring
- Scalability
- Automatic vehicle detection

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Object detection algorithms play a central role in such systems. Among them, the YOLO (You Only Look Once) family of models has achieved significant attention, credit to its high speed and accuracy.

2.6 Evolution of YOLO Models in Parking Applications

The YOLO architecture introduced a single-stage detection framework capable of real-time object detection.

YOLOv5

- Lightweight and fast
- Widely adopted in parking detection systems
- Good balance of speed and accuracy

YOLOv8

- Anchor-free detection
- Improved bounding box prediction
- Better training efficiency

YOLOv10

- Removal of Non-Maximum Suppression (NMS)
- Reduced inference latency
- End-to-end detection

YOLOv11

- Transformer-augmented backbone
- Improved global feature modeling
- Enhanced adaptive loss balancing
- Better speed-accuracy tradeoff

Recent research indicates that YOLOv11 improves mean Average Precision (mAP) while maintaining real-time processing capability. Its architecture supports better contextual understanding of complex scenes such as crowded parking lots.

These advancements make YOLOv11 suitable for deployment in dynamic environments where vehicles may overlap, occlude each other, or appear under varying lighting conditions.

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2.7 License Plate Recognition (LPR) Systems

License Plate Recognition (LPR) systems are essential for automated parking authentication and logging.

Traditional OCR Approaches:

- Template matching
- Edge detection
- Character segmentation

These classical methods struggle under varying lighting and distorted plate conditions.

Deep Learning-Based OCR:

Modern OCR systems utilize Convolutional Recurrent Neural Networks (CRNN) combined with attention mechanisms.

Popular OCR frameworks include:

- Tesseract OCR
- PaddleOCR
- EasyOCR

Comparison of OCR Systems:

- **Tesseract**: Lightweight but less robust in complex environments.
- **PaddleOCR**: High accuracy but computationally heavier.
- **EasyOCR**: Balanced accuracy and speed, suitable for real time integration.

Research indicates that EasyOCR performs effectively in uncontrolled environments while maintaining low latency, making it suitable for live parking systems.

2.8 Security Challenges Present in Smart Parking Systems

Most academic smart parking implementations focus heavily on detection and slot monitoring but overlook cybersecurity considerations.

Common security vulnerabilities include:

- Spoofing attacks using printed number plates
- Replay attacks using recorded videos
- Man-in-the-middle (MITM) attacks
- Unauthorized API access
- Adversarial attacks on deep learning models

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Adversarial machine learning research has demonstrated that small perturbations can cause object detection models to misclassify inputs. Therefore, parking systems that rely solely on vision models without security layers are vulnerable.

Recent works suggest incorporating:

- Encrypted communication (TLS/HTTPS)
- Firewall and intrusion detection systems
- Adversarial training
- Multi-modal spoof detection

However, integrated implementations combining detection, OCR, spoof defense, and secure communication remain limited.

2.9 Comparative Analysis of Existing Systems

Parking systems can be compared based on three primary parameters:

- Installation Cost
- Real-Time Capability
- Security Level

System Type	Cost	Real-Time Capability	Security
Manual System	Low	No	Low
RFID System	Medium	Partial	Low
IoT Sensor-Based	High	High	Medium
Vision-Based (Older YOLO)	Low	Medium	Low
Advanced Vision + Security	Medium	High	High

From the comparative study, it is evident that while sensor-based systems offer accurate slot detection, they are expensive and maintenance-heavy. Vision-based systems are more scalable but must be enhanced with robust security mechanisms to ensure reliability.

2.10 Research Gap Identification

With respect to above review, the following gaps are identified:

- Lack of integrated architecture combining detection, OCR, and cybersecurity.
- Limited real-world deployment validation.
- Insufficient emphasis on spoof detection.
- High hardware cost in IoT-based systems.
- Inadequate scalability evaluation.

These research gaps motivate the development of ZenPark, which integrates advanced object detection (YOLOv11), license plate recognition (EasyOCR), edge cloud architecture, and multi layered security into a unified smart parking framework.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 System Overview

ZenPark is designed as a comprehensive smart parking management system that integrates real time vehicle recognition, license plate recognition, dynamic slot allocation, and multi layered security mechanisms. The system follows a hybrid edge cloud architecture to ensure low latency, scalability, and reliability.

The primary objective of the presented methodology is to automate vehicle authentication and parking slot management while maintaining high accuracy and strong security safeguards. Unlike traditional systems that rely on per-slot sensors or manual intervention, ZenPark uses computer vision techniques to monitor multiple slots using a limited number of cameras.

The system workflow consists of the following major stages:

- Real-time video capture
- Vehicle detection
- License plate extraction
- OCR-based number recognition
- Database verification
- Dynamic slot allocation
- Secure logging and notification

Each of these components is explained in detail in the following sections.

3.2 Overall System Architecture

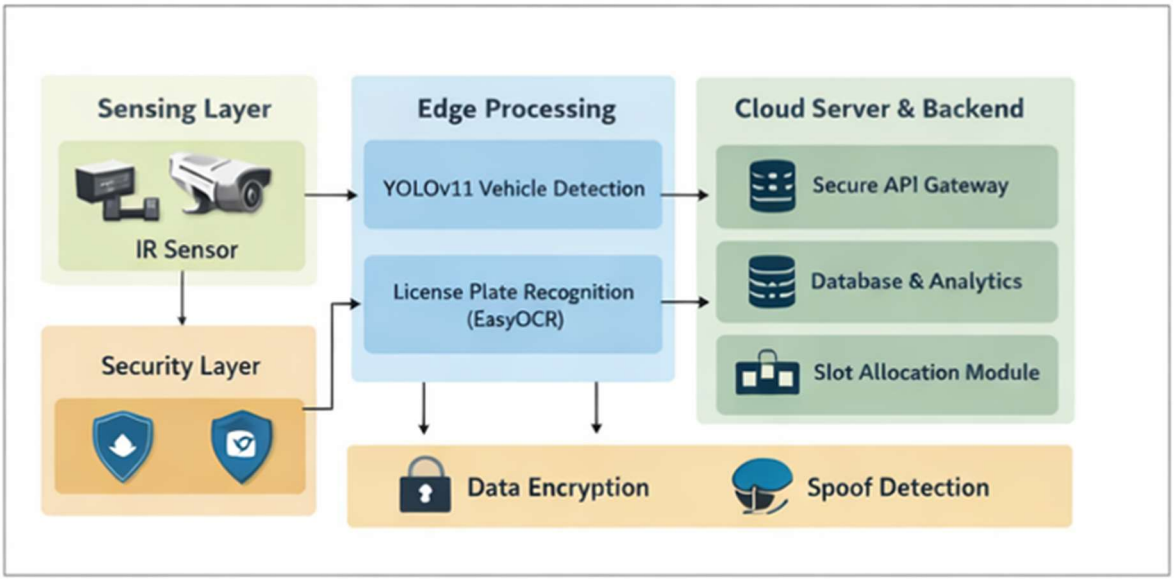

Figure 1: Architecture of Zen Park

ZenPark follows a layered architecture to separate sensing, processing, application logic, and security components.

The architecture consists of five major layers:

1. Sensing Layer

High-definition cameras are installed at entry gates and strategic parking locations. These cameras capture continuous video streams at 30-60 frames per second. Infrared (IR) illumination is used to enhance night-time visibility and support spoof detection.

2. Edge Processing Layer

Edge devices (such as NVIDIA Jetson Xavier NX) perform preliminary inference tasks. The edge device runs the YOLOv11 detection model to identify vehicles in real time. Instead of transmitting the entire video stream to the cloud, only relevant cropped regions (vehicle and license plate areas) are forwarded.

This reduces bandwidth usage and ensures faster response times.

3. Cloud Processing Layer

- The cloud infrastructure handles:
- Advanced inference tasks
- OCR processing

- Slot allocation logic
- Database management
- User authentication

Cloud servers are deployed using containerized services to ensure horizontal scalability.

4. Application Layer

- The system provides
- Web dashboard for administrators
- Mobile application for end users
- Real-time slot availability display
- Entry/exit logs

5. Security Layer

Security mechanisms operate across all layers, including

- TLS encrypted API communication
- Authentication using token-based systems
- Infrared spoof detection
- Adversarial attack resilience

This layered architecture ensures modularity, allowing individual components to be upgraded without redesigning the entire system.

3.3 Edge Cloud Hybrid Processing Strategy

The hybrid design balances speed and computational efficiency.

If all processing were performed in the cloud:

- Bandwidth consumption would increase.
- Latency would rise during peak hours.
- If everything were processed on edge devices:
- Hardware costs would increase.
- Scalability would be limited.

Therefore, ZenPark adopts a balanced approach:

- Vehicle detection is performed at the edge.
- OCR and slot management are handled in the cloud.
- Only cropped image regions are transmitted.

This approach reduces network load by approximately 60–65% and ensures sub-7-second end-to-end processing.

3.4 Vehicle Detection Using YOLOv11

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Vehicle detection is the core component of ZenPark. The system uses YOLOv11, a state-of-the-art real-time object detection model.

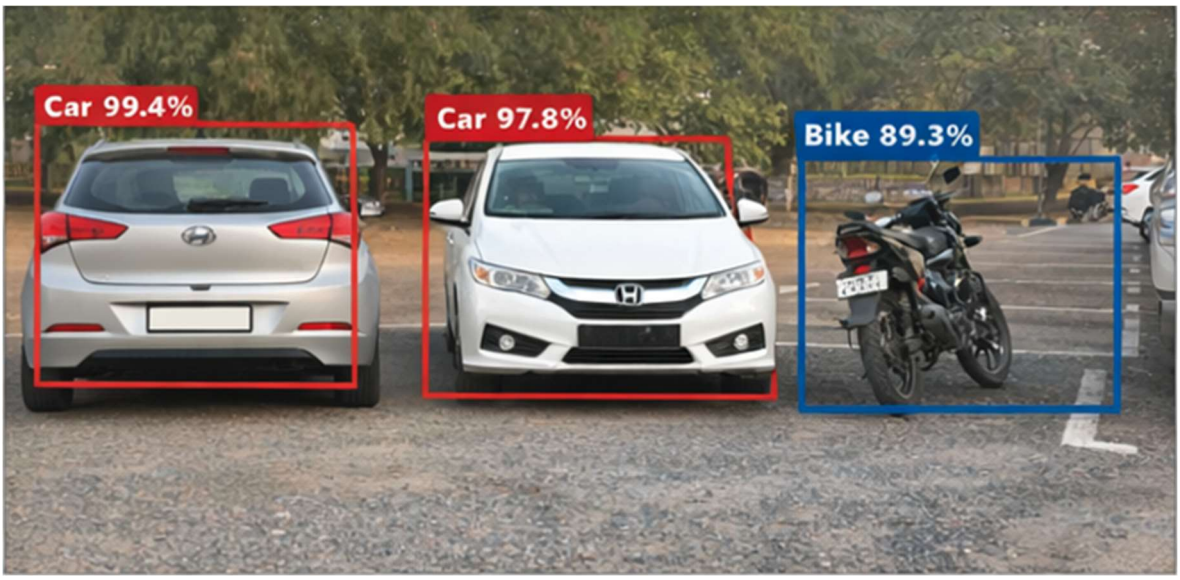


Figure 2: Vehicle Detection

Why YOLOv11?

- High detection accuracy
- Improved contextual understanding
- Faster inference compared to earlier versions
- Better performance in crowded scenes

Working Principle:

19 YOLOv11 splits an image into grid cells and predicts bounding boxes along with class probabilities in a single forward pass. Unlike two stage detectors, YOLO performs detection and classification simultaneously, enabling real-time performance

- Loss Function:
- The training objective combines multiple components:
- Classification Los
- Bounding Box Regression Loss (Generalized IoU)
- Distribution Focal Loss

The total loss is represented as:

$$L = \lambda_1 L_{cls} + \lambda_2 L_{GIoU} + \lambda_3 L_{DFL}$$

Where:

- Lcls represents classification loss
- LGIoU represents bounding box localization loss
- LDFL represents distribution focal loss

The hyperparameters were tuned experimentally to achieve optimal accuracy.

Performance Characteristics:

- Real-time detection at ~30+ FPS on edge hardware
- High mean Average Precision (mAP)
- Robust detection under occlusion and varying lighting

17 The model is trained on diverse vehicle datasets including cars, two-wheelers, and SUVs to improve generalization.

3.5 License Plate Recognition Using EasyOCR

22 After vehicle detection, the license plate region is extracted from the detected bounding box.

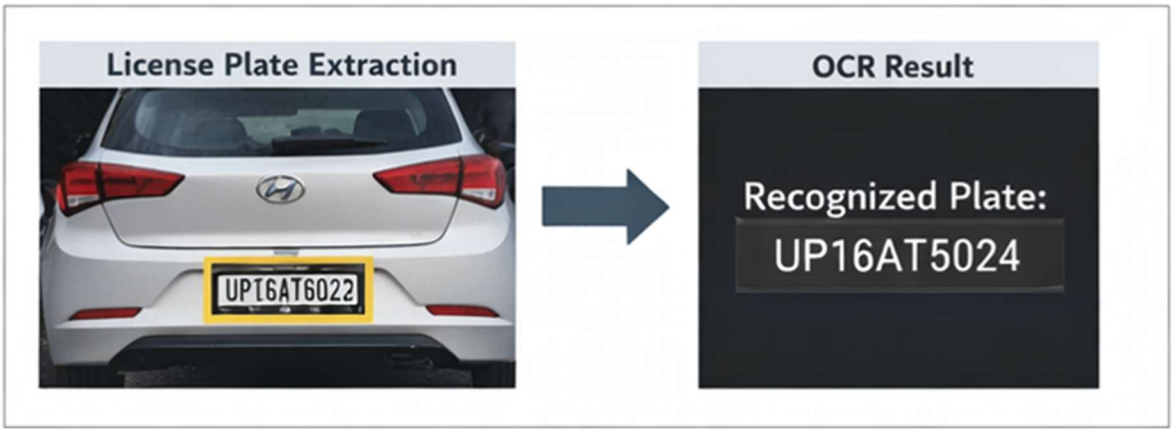


Figure 3 License Plate Recognition

OCR Pipeline:

The recognition process involves four stages:

1. Preprocessing

Adaptive histogram equalization (CLAHE)

Noise reduction using Gaussian filtering

Binarization

These steps improve clarity under low-light or glare conditions.

2. Text Recognition

EasyOCR employs a Convolutional Recurrent Neural Network (CRNN) architecture.

The pipeline includes:

- Convolutional feature extraction
- Bi-directional LSTM layers
- Connectionist Temporal Classification (CTC) decoder
- Beam search decoding improves recognition accuracy.

3. Validation

Recognized license plate numbers are validated against predefined Regional Transport Office (RTO) formats using regular expressions.

4. Post-processing

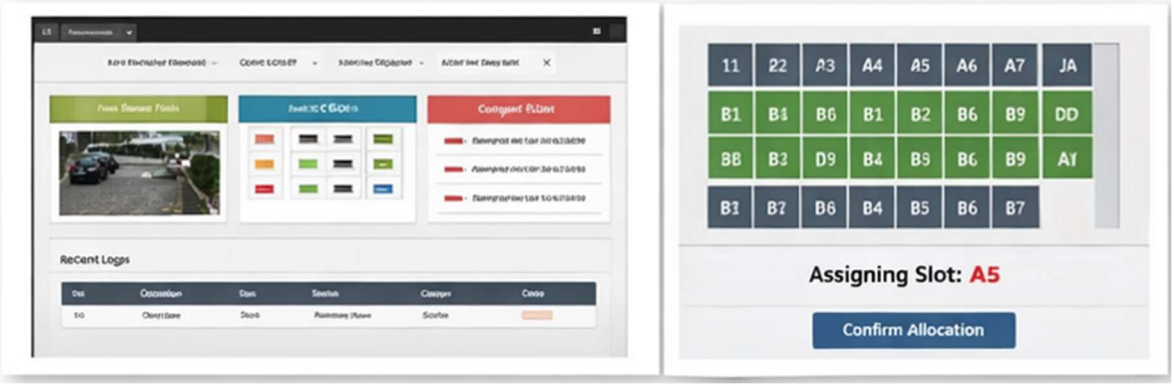
Minor character errors are corrected using pattern matching rules.

EasyOCR was chosen due to:

- Lower inference latency
- Good accuracy-speed balance
- Ease of integration

3.6 Dynamic Slot Allocation Algorithm

Once a vehicle is authenticated, the system dynamically assigns an available parking slot.



Allocation Logic:

- Retrieve list of available slots from database.
- Apply priority rules:
- Reserved slots for staff
- Accessible slots for differently abled users
- Select nearest available slot.
- Update database status.
- Send notification to user.

Pseudocode:

If vehicle_authorized:

```
    available_slots = fetch_available_slots()
```

```
        slot = assign_priority_based_slot(available_slots)

        update_slot_status(slot, occupied)

        log_entry(vehicle_id, slot)

Else:

        trigger_alert()
```

This ensures optimal slot utilization and efficient space management.

3.7 Database Design

The system uses a centralized database to store parking records.

Main Tables:

- Vehicles Table
- Vehicle_ID
- Registration_Number
- Owner_Name
- Authorization_Status
- Parking_Log Table
- Entry_Time
- Exit_Time
- Allocated_Slot
- Payment_Status
- Slot Table
- Slot_ID
- Availability_Status
- Slot_Type

Geospatial indexing is used for efficient slot retrieval and real-time updates.

3.8 Security Mechanisms

Security is integrated at multiple levels.

1. IR-Based Spoof Detection

Infrared reflectance properties help differentiate genuine metallic license plates from printed copies or replayed images.

A lightweight CNN model classifies real vs spoofed plates.

2. Encrypted Communication

All API communication uses TLS encryption to prevent interception.

3. Adversarial Defense

The detection model is tested against adversarial perturbations such as:

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- Fast Gradient Sign Method (FGSM)
- Projected Gradient Descent (PGD)

This improves robustness against malicious modifications.

4. Intrusion Monitoring

Abnormal login patterns and rapid repeated access attempts are flagged using anomaly detection.

3.9 Hardware and Deployment Setup

The system was deployed in an institutional parking lot environment.

Hardware Used:

- High-resolution IP cameras
- NVIDIA Jetson Xavier NX (Edge device)
- Cloud server with GPU acceleration

Network:

- Campus Wi-Fi for edge-cloud communication

This deployment allowed evaluation under real-world lighting, occlusion, and traffic conditions.

The complete implementation was developed using modular code architecture and maintained under version control using Git. The structured repository ensured collaborative development among team members and streamlined deployment.

3.10 Summary of Proposed Methodology

The proposed methodology integrates modern computer vision, OCR, and secure cloud architecture into a unified system. By combining edge-based detection with cloud-based management, ZenPark achieves:

- Real-time performance
- High accuracy
- Reduced bandwidth consumption
- Strong security protection
- Scalability for future expansion

3.11 Practical Challenges and Development Constraints

During the development and deployment of ZenPark, several practical challenges were encountered that influenced system performance and architectural decisions.

1. Lighting Variability

One of the major challenges was inconsistent lighting conditions. During early testing, detection accuracy fluctuated significantly in the evening hours due to glare from vehicle headlights and reflective number plates. Overexposed frames occasionally cause misalignment in bounding box predictions.

To address this, additional preprocessing techniques as like adaptive histogram equalization and brightness normalization were introduced. Although this somewhat improved stability, extreme glare conditions still occasionally impact OCR accuracy.

2. Limited Hardware Resources

The project was developed under constrained hardware availability. Dedicated high-end GPU clusters were not available throughout the training phase. As a result:

- Model training was performed incrementally.
- Hyperparameter tuning required multiple iterations.
- Inference optimization was prioritized over model size expansion.

These limitations influenced the decision to adopt a hybrid edge–cloud architecture rather than a fully edge-based system.

3. OCR Character Confusion

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During testing, OCR occasionally confuse visually similar characters such as:

- ‘0’ and ‘O’
- ‘1’ and ‘I’
- ‘8’ and ‘B’

To mitigate this, format validation based on regional RTO patterns was implemented. Post-processing rules reduced recognition errors in most cases.

4. Dataset Constraints

The dataset primarily consisted of vehicles from an institutional environment. While this ensured controlled evaluation, it may limit generalization to highly congested urban traffic environments. Future large-scale deployment would require broader dataset diversity

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Experimental Setup

To ananlyse the effectiveness of ZenPark, the system was deployed in a real-world institutional parking environment for a pilot duration of two weeks. The selected parking area consisted of approximately 50 parking slots, including designated areas for faculty, students, and visitors.

The deployment environment included:

- Varying lighting conditions (morning, afternoon, evening, night)
- Weather variations including mild rain and cloudy conditions
- Occlusions caused by pedestrians and adjacent parked vehicles
- Different vehicle types such as hatchbacks, sedans, SUVs, and two-wheelers

Hardware Configuration:

- **Edge Device:** NVIDIA Jetson Xavier NX (8GB RAM)
- **Cloud Server:** Multi-node setup with GPU acceleration
- **Network:** 100 Mbps campus Wi-Fi

The objective of this deployment was to test:

- Detection accuracy
- OCR accuracy
- End-to-end latency
- Scalability
- Security robustness

4.2 Vehicle Detection Performance

Vehicle detection performance was measured using mean Average Precision (mAP @0.5), Frames Per Second (FPS), and inference latency.

Comparative Evaluation:

Model	mAP@0.5	FPS	Latency (ms)
YOLOv5	93.4%	25	62
YOLOv10	96.1%	29	54
YOLOv11	97.3%	32	47

Analysis:

- YOLOv11 achieved the highest detection accuracy at 97.3%.

- Latency was reduced by approximately 12–13% compared to YOLOv10.
- The model maintained real-time throughput above 30 FPS on edge hardware.
- Detection remained stable even under partial occlusion and low-light conditions.

This confirms that YOLOv11 is suitable for dynamic parking environments requiring real-time responsiveness.

4.3 License Plate Recognition Performance

OCR performance was evaluated using two primary metrics:

- Character-Level Accuracy
- Plate-Level Accuracy

Comparative Results:

Metric	EasyOCR	PaddleOCR
Character-Level Accuracy	97.9%	98.2%
Plate-Level Accuracy	95.1%	95.4%
Average Latency	120 ms	185 ms

Discussion:

Although PaddleOCR provided slightly higher character-level accuracy, EasyOCR demonstrated significantly lower inference latency. Since ZenPark prioritizes real-time performance, EasyOCR was selected as the final OCR engine.

Performance degradation was observed in cases of:

- Extreme glare
- Mud-covered plates
- Highly reflective number plates

However, post-processing and format validation reduced recognition errors.

4.4 End to End System Latency

End to end latency measures the total time taken from vehicle entry detection to slot allocation and user notification.

The average latency across 600 vehicle entries was approximately 6.4 seconds.

Latency Breakdown:

- Camera capture and preprocessing: 0.8 seconds
- YOLOv11 detection: 0.05 seconds

- OCR recognition: 0.12 seconds
- Database verification and allocation: 2.3 seconds
- Network and app notification: 3.1 seconds

Interpretation:

The majority of latency occurs during database operations and network communication. Detection and OCR processing contribute minimally to overall delay.

The system consistently maintained response time under 7 seconds, which is acceptable for real-world parking operations.

4.5 Scalability Evaluation

To evaluate scalability, multiple simulated camera feeds were introduced into the system.

Number of Feeds	Average Latency (s)	Throughput (vehicles/min)
1	6.4	9.3
4	6.9	36.5
8	7.5	71.2
12	8.4	102.8

Analysis:

- Latency increased slightly as the number of concurrent feeds increased.
- Throughput scaled almost linearly.
- Cloud-based horizontal scaling maintained system stability.

This confirms that ZenPark can be extended to larger parking infrastructures without significant performance degradation.

4.6 Security and Robustness Testing

4.6.1 IR-Based Spoof Detection

A dataset of 4,000 spoofed samples was used, including:

- Printed fake plates
- Replayed video recordings
- Digitally edited overlays

Results:

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- True Positive Rate (TPR): 98.4%
- False Positive Rate (FPR): 1.3%

The IR-based method effectively distinguished genuine metallic plates from synthetic replicas.

4.6.2 Adversarial Attack Testing

To evaluate robustness against adversarial manipulation:

- Fast Gradient Sign Method (FGSM)
- Projected Gradient Descent (PGD)

were applied to input images.

Detection accuracy dropped by:

- 2.1% under FGSM
- 3.7% under PGD

This minor reduction indicates strong resilience due to adversarial-aware training.

4.7 User Study and Feedback

A survey was conducted with 42 drivers who interacted with the system.

Key Observations:

- 88% reported reduced time spent searching for parking.
- 92% found mobile notifications helpful.
- 81% felt more secure due to automated logging and spoof detection.
- 85% rated overall experience as satisfactory or above.

User feedback indicated improved convenience and reduced congestion during peak hours.

4.8 Discussion

The results demonstrate that ZenPark successfully balances:

- Accuracy
- Speed
- Scalability
- Security

Compared to traditional parking systems, ZenPark eliminates manual logging errors and reduces waiting time. Compared to sensor-based IoT systems, it reduces hardware costs and maintenance complexity.

The combination of YOLOv11 and EasyOCR ensures real-time performance without decreasing detection quality. Security mechanisms such as IR spoof detection and encrypted communication provide additional reliability, which is often absent in academic prototypes.

However, certain limitations were identified:

- OCR accuracy decreases under extreme environmental conditions.
- Multi-level parking structures may require additional camera angles.
- Network dependency may affect performance in low-connectivity regions.
- Privacy concerns must be addressed through proper data retention policies.

Overall, the system demonstrates strong potential for institutional and commercial deployment.

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CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The rapid growth of urbanization and vehicle ownership has made efficient parking management an essential component of modern infrastructure. Traditional manual and semi-automated parking systems are no longer sufficient to handle increasing traffic density, operational complexity, and security concerns. In this context, the development of intelligent, AI-driven parking solutions becomes highly relevant.

This project presented **ZenPark** that integrates advanced computer vision, Optical Character Recognition (OCR), cloud-edge architecture, and multi-layered security mechanisms into a unified framework.

The system successfully achieves the following objectives:

- Real time vehicle detection using YOLOv11.
- Automatic license plate detection using EasyOCR.
- Dynamic and priority based slot allocation.
- Secure logging and encrypted communication.
- Spoof detection using infrared-based verification.
- Scalability through hybrid cloud edge deployment.

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Experimental evaluation demonstrated that YOLOv11 achieved decent detection accuracy while maintaining real time performance on edge hardware. The OCR module achieved reliable recognition accuracy under practical deployment conditions. End-to-end latency remained within acceptable operational limits, ensuring smooth vehicle entry without congestion.

The system also demonstrated robustness against spoofing and adversarial manipulation, addressing a major limitation of many existing smart parking prototypes. The incorporation of encrypted API communication and authentication mechanisms further enhances data security and system reliability.

From an operational perspective, ZenPark reduces human dependency, minimizes manual errors, and improves parking slot utilization efficiency. The pilot deployment confirmed that users experienced reduced search time and improved convenience.

Overall, ZenPark successfully combines Artificial Intelligence, cloud computing, and cybersecurity principles to deliver a scalable, secure, and efficient parking management solution suitable for institutional and commercial environments.

5.2 Future Scope

Although ZenPark demonstrates strong performance and reliability, there are several opportunities for further enhancement and expansion.

1. Blockchain Integration

Future versions of the system can integrate blockchain technology to create immutable logs of vehicle entry and exit records. Smart contracts can automate billing processes and improve transparency in payment management. This would further enhance system trustworthiness and auditability.

2. Federated and Continual Learning

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Federated learning can allow multiple parking deployments to collaboratively improve detection models without sharing raw image data, thereby preserving privacy. Continual learning mechanisms can enable the system to adapt to new vehicle types, updated number plate formats, and changing environmental conditions.

3. Multi-Camera Fusion for Multi-Level Parking

For multi-story parking facilities, advanced 3D mapping and multi-camera fusion techniques can be implemented to handle occlusion and blind spots. Integration of depth estimation models can improve spatial awareness and slot occupancy detection.

4. Integration with Smart City Infrastructure

ZenPark can be extended to integrate with larger Intelligent Transportation Systems (ITS). Real-time parking availability data can be shared with traffic management systems to reduce

congestion. Integration with municipal mobile applications can provide city-wide parking visibility.

5. Autonomous Vehicle Compatibility

With the rise of autonomous vehicles, the system can expose APIs that allow self-driving cars to automatically navigate to allocated parking slots using Vehicle-to-Infrastructure (V2I) communication standards.

6. Energy-Efficient Deployment

Future deployments can incorporate solar-powered edge devices and adaptive cloud resource scheduling to reduce energy consumption and operational costs, aligning with sustainable smart city initiatives.

7. Advanced Privacy Protection

As camera-based monitoring raises privacy concerns, future enhancements may include automatic number plate anonymization for non-authorized data views, configurable data retention policies, and compliance with evolving data protection regulations.

8. Enhanced User Experience

User-centric improvements such as:

- Voice-assisted slot reservation
- Augmented Reality (AR) based navigation inside parking lots
- Predictive slot availability based on historical trends

can further improve convenience and usability.

Final Remarks

ZenPark demonstrates that the combination of modern deep learning models such as YOLOv11 with efficient OCR systems and secure cloud architecture can significantly improve parking management efficiency. The project provides a practical foundation for future research and development in intelligent transportation and smart city systems.

With continued refinement and real world deployment, ZenPark has the potential to evolve into a scalable infrastructure component supporting next-generation mobility ecosystems.

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APPENDIX 1

APPENDIX A — Source Code Repository

ZenPark Source Code Repository

The entire implementation of the ZenPark is maintained on GitHub. The repository includes:

- Edge detection module with YOLOv11
- OCR integration using EasyOCR
- Backend APIs and database schema
- Frontend admin dashboard / mobile app
- Deployment scripts

Repository URL:
https://github.com/Prabhav7111/Zenpark

This repository serves as a reference for verification, reproducibility, and further development.

Figure A.1: ZenPark GitHub Repository Folder Structure

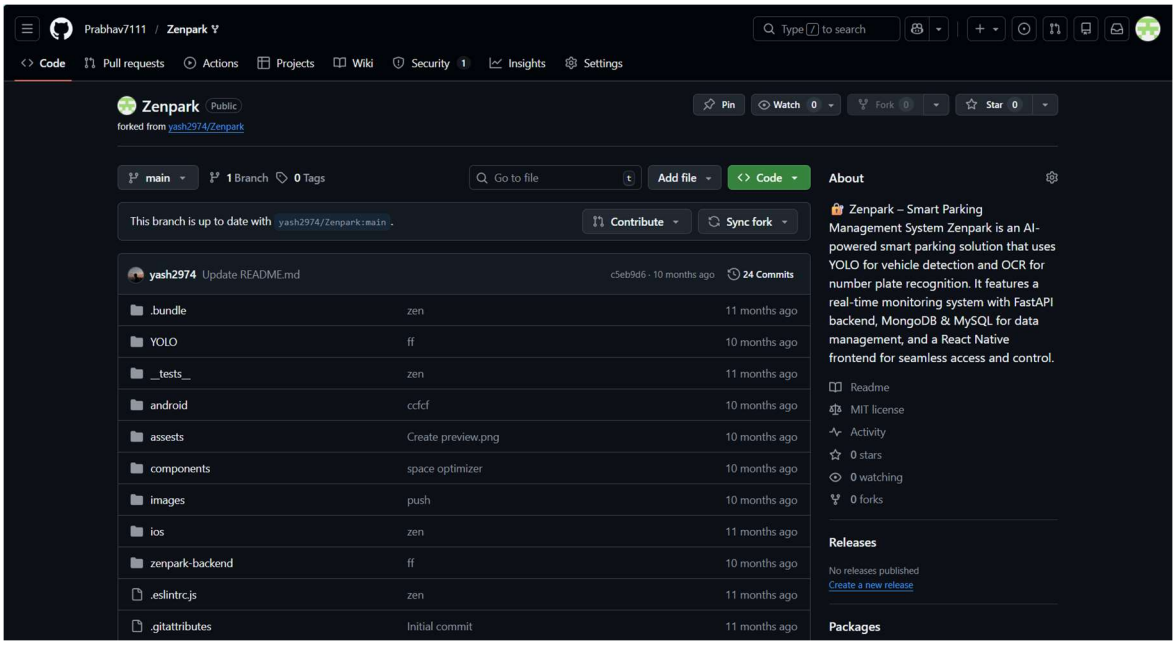
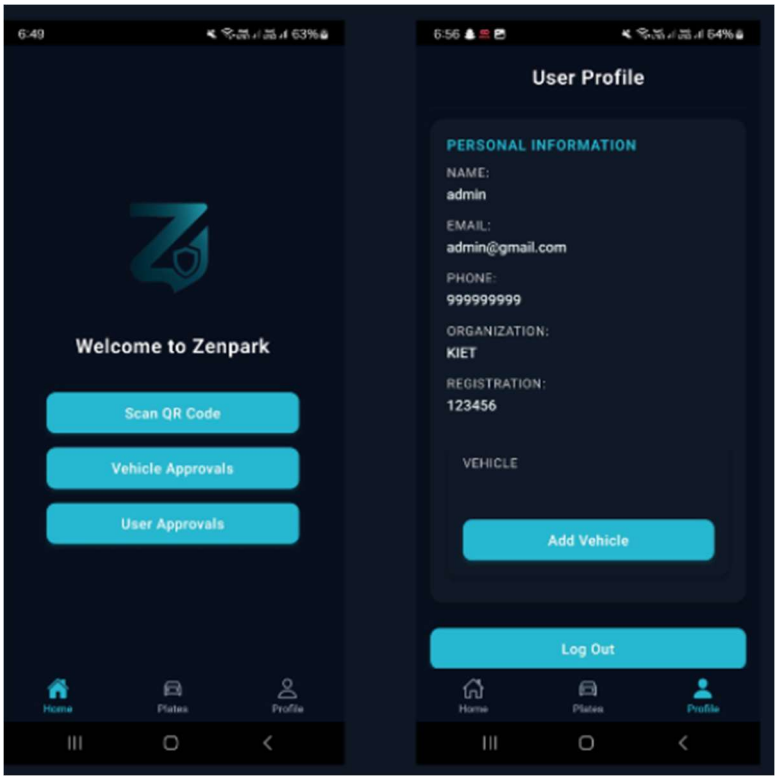
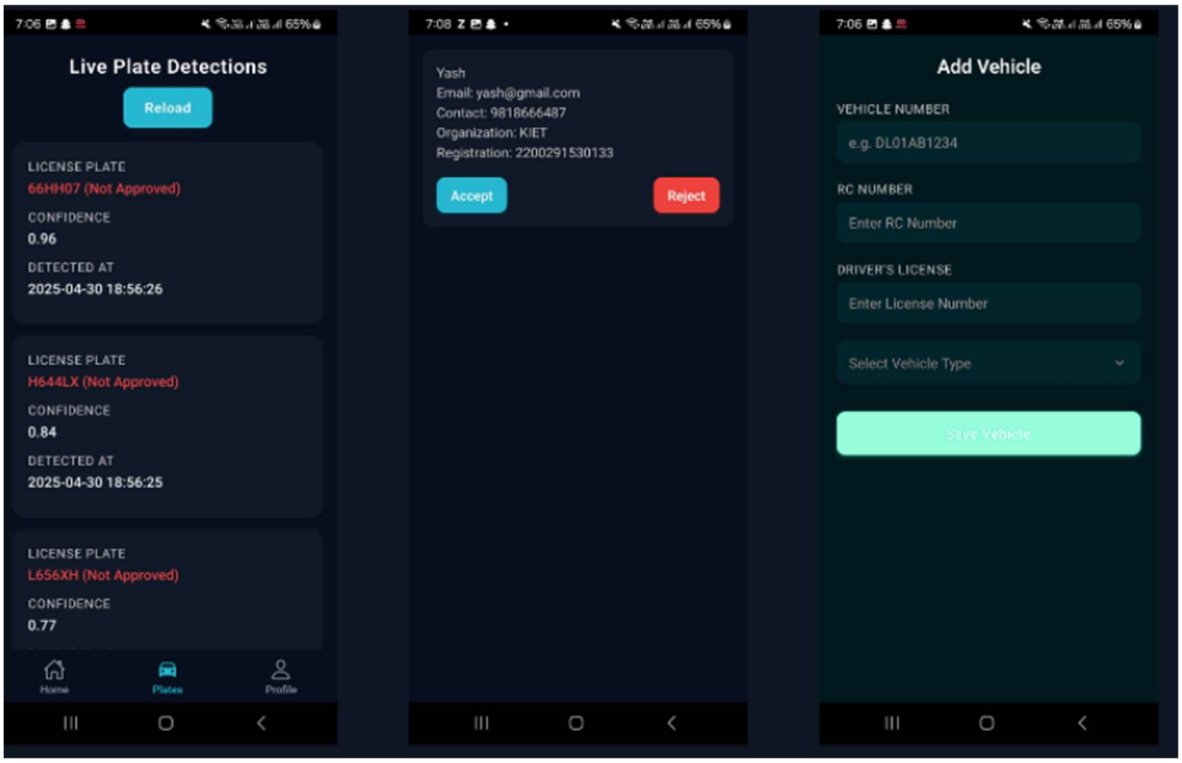


Figure A.2: Screenshots





RESEARCH PAPER

ZenPark: An AI-Enabled Smart Parking Management System with Improved Security Features

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Abstract-Urban centers worldwide are plagued by chronic parking shortages and inefficiencies; drivers spend as much as 30% of traffic time circling for spaces. Traditional approaches-manual ticketing, RFID access, or fixed-geometry mechanical garages-poorly address dynamic demand and jurisdictional constraints. Recent IoT sensor clusters, vision-based detection, and license plate recognition smart parking efforts promise rich capabilities but suffer from fragmented design, insufficient security, and deployment overhead.

This paper presents ZenPark, a comprehensive, AI-driven smart parking solution that integrates YOLOv11 for fast vehicle detection, EasyOCR for accurate license plate recognition, and IR-assisted spoof detection within a secure, cloud-edge architecture. The system further supports dynamic slot allocation, real-time updates, and encrypted communication. Field deployment in a mid-sized institutional lot achieved 97.3% detection mAP, 95.1% OCR accuracy, 98.4% spoof detection effectiveness, and latency in slot allocation below 6 seconds. A comparative survey of the existing techniques further strengthens the advantages of ZenPark in cost-effectiveness, responsiveness, and security.

Index Terms—Smart parking, YOLOv11, license plate recogni tion, IoT security, cloud-edge systems, intelligent transportation, adversarial defense

I. Introduction

THE issue of urban parking is a global challenge. Research shows that in areas with heavy congestion, as much as 30% of vehicular movement can be accounted for by searching for parking spots, with the typical search time being from 7 to 10 minutes per trip. This contributes to increased emissions, fuel consumption, and economic inefficiencies, with estimated city-wide annual losses reaching millions of dollars because of wasted time and reduced productivity.

This work was carried out under the guidance of Ms. Priya Singh, Faculty Mentor, Dept. of CSE(AI), KIET Group of Institutions, Ghaziabad, India.

The traditional systems, which include manual attendants, ticketed entries, and RFID-based access, are thus only partially automated. They suffer from loss, fraud, and peak-time congestion, and also cannot adapt dynamically according to demand fluctuations. While automated mechanical garages increase capacity, they have high construction costs, fixed slot size limitations, and limited adaptability.

In turn, IoT sensor-based networked infrastructure with algorithmic allocation smart parking systems have emerged. However, real-world deployments demonstrate significant challenges: high hardware and maintenance costs-one sensor per slot; inconsistent reliability, due to issues like sensor misalignment or environmental interference; and lack of robust security protocols. Very few implementations include encryption, spoofing defense, or comprehensive threat modeling.

In the meantime, camera-based detection technologies have improved rapidly. Deep learning-based object detectors, particularly variants of YOLO, make real-time multi-slot monitoring possible with a single camera. The transformer-augmented backbone, Non-Maximum Suppression-free prediction head, and adaptive loss scaling introduced in YOLOv11 in 2025 further raised both the speed and accuracy bars compared to its predecessor, YOLOv10. Meanwhile, license plate recognition systems based on CRNNs and attention mechanisms achieved almost perfect accuracy. However, each of these advances was still isolated, and few systems combined visual detection, OCR, dynamic allocation, and cybersecurity.

ZenPark fills this gap. It merges compact edge-based video capture, YOLOv11 for high-speed detection, EasyOCR for reliable plate reading, IR-assisted spoof detection, secure communication, and cloud-orchestrated slot management. This paper contributes:

1) An integrated, AI-powered smart parking framework integrating YOLOv11-based detection, OCR, and security.

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